

Linux & SQL Proficiency

Google Cybersecurity Professional Certificate Portfolio

Introduction

This portfolio project highlights foundational cybersecurity tools and technical skills involving Linux operating systems and SQL database management. These technologies are essential tools of trade used by cybersecurity professionals, system administrators, and security analysts.

Linux Proficiency

Linux is widely used in cybersecurity because of its flexibility, security features, and command-line capabilities. Through practical exercises, I developed proficiency in navigating Linux environments, managing files and directories, monitoring system performance, and using networking utilities.

Key Linux Skills Demonstrated

- Command-line navigation and file management
- User and permission management
- Network monitoring and troubleshooting
- System process analysis
- Use of administrative commands with sudo privileges
- Log inspection and basic security auditing

SQL Proficiency

Structured Query Language (SQL) is an important skill in cybersecurity and data management. SQL allows professionals to retrieve, analyze, and manage data stored in relational databases. During this training, I practiced writing SQL queries to search, filter, and organize information.

Key SQL Skills Demonstrated

- Writing SELECT queries
- Filtering records using WHERE clauses
- Sorting data using ORDER BY
- Using aggregate functions such as COUNT and AVG
- Understanding database structures and relationships

Cybersecurity Relevance

Linux and SQL skills are essential in cybersecurity operations because security analysts often work with Linux servers and large datasets. These tools support threat analysis, incident response, system monitoring, and vulnerability management.

Conclusion

This project demonstrates foundational proficiency in Linux and SQL as critical tools of trade within cybersecurity. These technical skills provide the ability to manage systems, analyze data, and support security operations effectively.